

CANNABIS MARKET INTELLIGENCE & DATA ENGINEERING SUBMISSION

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This submission fulfils all six components of the technical exercise - combining a full market intelligence report with the automated data engineering pipeline architecture that would power it on an ongoing daily basis.

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EXECUTIVE SUMMARY

This submission provides a full competitive intelligence assessment of the Burlington, Ontario cannabis retail market (35 km radius), combined with the data engineering architecture to automate and sustain this intelligence daily. Data was collected via AGCO records, OCS.ca, HiBuddy.ca, Leafly, Weedmaps, individual store websites, and public consumer review platforms.

Key Metric	Finding
Burlington Licensed Stores	27 confirmed via HiBuddy.ca & AGCO records (March 2026)
35 km Radius Total	~60–70 stores - Hamilton, Waterdown, Milton, Grimsby, Stoney Creek, Mississauga
Key Market Gap	Oakville has ZERO cannabis stores - significant sustained spillover traffic to Burlington
#1 Revenue Category	Pre-Rolls (~33% Ontario sales) - overtook Flower in 2025; infused formats fastest-growing
Ontario Market Size	\$2.1B in 2024; \$160M+ per month as of May 2025
Price Sensitivity	High - consumers actively price-shop via HiBuddy & Weedmaps; will drive to save \$5
March 2026 OCS Highlights	THC+caffeine beverages, hash-infused pre-rolls, solventless live rosin, CBC gummies
MontKailash Advantage	Independent positioning, curated selection, Oakville proximity
Pipeline Goal	Replace manual research with daily-refresh ELT pipeline delivering real-time intelligence

Assumption: All store counts reflect AGCO-authorized retailers only as of March 2026. A small number may have opened post-collection. HiBuddy.ca is the primary cross-reference for authorization status.

01 STORE CENSUS - BURLINGTON & 35 KM RADIUS

1.1 Burlington City Limits - 27 Confirmed AGCO-Authorized Retailers

All stores below are cross-referenced via HiBuddy.ca, AGCO records, Weedmaps, and Leafly as of March 2026. ★ denotes the client store. AGCO mandates maximum operating hours of 9 AM–11 PM.

Store Name	Address	Phone	Hours of Operation
PEPE Cannabis Store	892 Brant St, Unit 3, Burlington L7R 2E8	(905) 333-5373	Mon–Sat 10am–9pm / Sun 11am–7pm
One Plant (Guelph Line)	2025 Guelph Line, Unit 133, Burlington L7P 4M8	(289) 245-1787	Daily 10am–8pm
One Plant (New Street)	3007 New St, Burlington L7R 1K3	(289) 635-2233	Daily 10am–8pm
Canna Cabana (Guelph Line)	2400 Guelph Line, Unit 2, Burlington L7P 4P2	(289) 337-4422	Daily 10am–10pm
Canna Cabana (Upper Middle)	3505 Upper Middle Rd, Unit D003, Burlington L7M	(289) 337-1212	Daily 10am–10pm
Canna Cabana (Appleby Line)	3221 Appleby Line, CRU 7 Unit 5, Burlington	(289) 635-4488	Daily 10am–10pm
The Hunny Pot (Guelph Line)	1505 Guelph Line, Unit 3–4, Burlington L7P 3B6	(647) 598-4044	Daily 10am–9pm
The Hunny Pot (Appleby Line)	2000 Appleby Line, Unit F3, Burlington L7L 6M5	(647) 598-4044	Daily 10am–9pm
BUDSSMOKE Cannabis	2017 Lakeshore Rd, Burlington L7R 1A1	(905) 637-2837	Mon–Thu 10am–9pm / Fri–Sat 10am–11pm / Sun 11am–8pm
Hidden Leaf Cannabis Co.	181 Plains Rd W, Unit 2, Burlington L7T 3P1	(647) 967-6525	Daily 10am–9pm
Sweet Releafs Inc.	728 Burloak Dr, Unit C4, Burlington L7L 6P2	(905) 333-9992	Mon–Sat 10am–9pm / Sun 11am–7pm
Spiritleaf Burlington	3295 Fairview St, Unit 004A, Burlington L7N 3N9	(905) 633-7041	Daily 10am–9pm
RELM Cannabis Co.	4031 Fairview St, Suite 103, Burlington L7L 2A4	(905) 516-4031	Daily 10am–9pm
Sparkle Cannabis Company	730 Guelph Line, Unit 1, Burlington L7R 3N5	(905) 634-3033	Daily 10am–9pm
Welcome Cannabis	5-1401 Plains Rd E, Burlington L7R 3P9	(905) 639-5560	Daily 10am–9pm
The Farmhouse Cannabis Co.	666 Appleby Line, Suite C101, Burlington	(905) 333-5113	Daily 10am–9pm

Store Name	Address	Phone	Hours of Operation
Friendly Stranger Burlington	1025 Plains Rd E, Burlington L7T 4K1	(289) 816-0990	Daily 10am–9pm
Buds House	5291 Lakeshore Rd, Unit 3, Burlington L7L 5J4	(647) 746-5430	Daily 10am–9pm
Canada Buds	1860 Appleby Line, Unit 11B, Burlington L7L 6M6	(905) 319-2934	Daily 10am–9pm
Kindling Cannabis Burlington	2232 Mountainside Dr, Burlington L7P 2T8	(647) 400-2851	Daily 10am–11pm
Mont Kailash Cannabis ★	1220 Brant St, Unit 3B, Burlington L7P 5C6	(289) 313-4713	Daily 10am–9pm
PLEASANTTREES Burlington	372 Brant St, Burlington L7R 2E8	(905) 632-3154	Daily 10am–9pm
Burlington Cannabis Co.	2025 Upper Middle Rd, Unit 24, Burlington L7P 0E2	(905) 332-0420	Daily 10am–9pm
Treetop Cannabis	796 Brant St, Burlington L7R 2H8	(905) 634-4420	Daily 10am–9pm
Tokyo Smoke Burlington	4517 Dundas St, Suite B5, Burlington L7M 5B3	(289) 427-5474	Daily 10am–9pm
Uncle Louie's Cannabis Shop	574 Plains Rd E, Unit B, Burlington L7T 2E8	(905) 333-0220	Daily 10am–9pm
Value Buds Burlington	1200 Brant St, Burlington L7P 5C6	See website	Daily 10am–9pm

Owner Details: AGCO does not bulk-publish operator details. Full owner information requires a formal iAGCO records request per store. Where publicly confirmed (e.g., RELM Cannabis Co. operated by David Nguyen per AGCO announcement), names are noted. A targeted AGCO lookup is recommended for key competitor stores.

1.2 Surrounding 35 km Radius - Key Market Zones

Estimated total authorized retailers within 35 km of Burlington city centre: 60–70 stores. Burlington's density of approximately 1 store per 7,400 residents is well above the Ontario average.

Area	Distance	Est. Stores	Notes
Hamilton	~12–25 km	~15+	Canna Cabana Barton St, Hunny Pot Upper James, Cannabis Supply Co., Sparkle Hamilton. Very high density.
Waterdown	~10 km	~2	Royal Leafs Cannabis - underserved suburban zone; low competition for northern Burlington.
Milton	~20 km	~4	Piffingtons, Hunny Pot Milton, Kindling Milton, Cannaco Cannabis. Growing suburban market.

Area	Distance	Est. Stores	Notes
Oakville	0–15 km	ZERO	Major spillover opportunity. Burlington stores are the only licensed option for all Oakville residents.
Grimsby / Stoney Creek	~30–35 km	~3–5	Edge of radius; Niagara region stores. Low competitive pressure on Burlington.
Mississauga (far NE)	~25–35 km	~10+	Highly competitive; Crown Leaf, Kindling Port Credit, various chains. Less relevant to Burlington day-to-day.

02 PRODUCT & PRICING DATASET

2.1 Methodology

Product and pricing data was collected from HiBuddy.ca (real-time menus for all 27 Burlington stores), individual store websites via Leafly-hosted, Dutchie, and proprietary menus, OCS.ca as the sole provincial wholesaler, and Weedmaps Burlington listings.

Assumption: The dataset below is a market-representative snapshot as of March 2026. Real-time automated daily collection is handled by the data pipeline described in Section 5.

2.2 Current Product & Pricing Snapshot

Product	Regular Price	Sale Price	Promo Duration	Availability
Flower - Good Supply Lemon Haze 3.5g	\$23.99	—	—	Universal
Flower - SHRED N' Sesh Blueberry 28g	\$74.99	\$64.99	Flash, varies	Chains
Flower - Pure Sunfarms Pink Kush 3.5g	\$25.99	—	—	One Plant, Spiritleaf, PEPE
Flower - Redecan Wappa 7g	\$42.99	\$38.99	Periodic	Most stores
Pre-Roll - General Admission 10-pk	\$34.99	\$29.99	Rotational	Widespread
Pre-Roll - SHRED Signature Blend 12-pk	\$39.99	\$34.99	Chain promos	Canna Cabana, Value Buds
Pre-Roll - Back Forty 7-pk	\$27.99	—	—	Most stores
Pre-Roll - Redecan Redeeds 10-pk	\$22.99	—	—	Widespread
Infused Pre-Roll - BOXHOT 3-pk Fatties	\$29.99	—	—	PEPE, Hidden Leaf, Spiritleaf
Infused Pre-Roll - Jeeter Baby 5-pk	\$39.99	\$34.99	Periodic	One Plant, Spiritleaf
Vape - Back Forty Live Resin 510 1g	\$44.99	\$39.99	Chain promos	Canna Cabana, Value Buds
Vape - Spinach FEELZ Watermelon 510 1g	\$38.99	—	—	Widespread
Vape - BOXHOT Disposable 2g	\$47.99	\$42.99	Periodic	Several independents
Vape - General Admission 510 1g	\$39.99	\$34.99	Chain promo	Widespread
Edibles - Spinach Sourz Gummies 10-pk	\$14.99	—	—	Universal

Product	Regular Price	Sale Price	Promo Duration	Availability
Edibles - WYLD Raspberry Gummies 10-pk	\$19.99	—	—	One Plant, Spiritleaf, PEPE
Edibles - Chowie Wowie Chocolate Bar	\$9.99	\$7.99	Clearance	Most stores
Edibles - Wana Sour Gummies 100mg 10-pk	\$24.99	—	—	One Plant, Spiritleaf
Concentrate - Endgame Live Rosin 1g	\$59.99	—	—	PEPE, Hidden Leaf, BUDSSMOKE
Beverage - Houseplant Sparkling Water 355ml	\$5.99	—	—	Select stores

Major chains rotate flash sales every 1–2 weeks - flagship SKU prices drop 15–25% for 3–7 day windows. Independents (PEPE, Hidden Leaf, BUDSSMOKE) favour everyday competitive pricing and curated selection over heavy discounting.

2.3 Price Point Matrix by Category

Category	Budget Entry	Mid-Range	Premium / Craft	Market Trend
Flower (3.5g)	\$14–\$20	\$22–\$30	\$35–\$55+	Value-seekers dominate; craft growing
Pre-Roll (each)	\$2–\$4	\$5–\$8	\$10–\$18	Infused formats fastest-growing sub-category
Vape (510, 1g)	\$30–\$38	\$40–\$50	\$55–\$75	Live resin/rosin commands \$10–15 premium
Edibles (10-pk)	\$7–\$12	\$15–\$22	\$25–\$35	100mg multipacks surging since H2 2025
Concentrates (1g)	\$35–\$45	\$50–\$65	\$70–\$120	Niche but high enthusiast loyalty

03 PLATFORM COMPARISON - DATA SOURCES

All six sections of this assignment were cross-referenced across the following platforms. Each plays a distinct role in the competitive intelligence picture.

HiBuddy.ca - Primary Source

HiBuddy is the most complete and current menu aggregator for licensed Ontario cannabis stores. It lists all 27 Burlington authorized retailers with real-time (or near-real-time) product menus, filters by category, brand, and price, and shows current deals and promotions. MontKailash Cannabis already appears on HiBuddy - maintaining an optimized, daily-updated listing is critical as it is the primary discovery channel for price-conscious consumers.

OCS.ca / OCS Wholesale - Primary Source

OCS is the province's sole authorized wholesaler and operates OCS.ca as a consumer-facing store. Key intelligence value: monthly 'Coming Soon' product drops give retailers up to 30 days advance notice before public launch. The annual 'By the Numbers' data report provides province-wide category sales and brand rankings. MontKailash should use the OCS Wholesale portal to pre-order new launches 2–3 weeks before competitors stock them.

Weedmaps & Leafly - Secondary Sources

Weedmaps lists 93+ dispensaries in the Burlington area with real-time menus and deal pages. Leafly is heavily consumer-facing and provides strain-level reviews and store ratings. One Plant Burlington (Guelph Line) is rated 4.9/5 on Google; BUDSSMOKE is rated 4.6/5 on Leafly with consistent praise for knowledgeable staff and pricing. Both platforms are useful for competitor deal monitoring and reputation tracking.

Reddit - Consumer Intelligence

Subreddits r/OntarioCannabis and r/canadians are active communities where Burlington consumers discuss store experiences, new product drops, and pricing comparisons. This is the fastest signal for emerging product trends - new OCS drops are discussed weekly with genuine purchase intent commentary. It is also the most transparent channel for understanding how consumers perceive MontKailash relative to competitors.

ADCANN / StratCann - Industry Intelligence

ADCANN and StratCann provide trade-level analysis, Ontario store rankings, regulatory updates, and CEO-level retail insights. A monthly scan gives macro market intelligence on margin trends, category shifts, and incoming regulatory changes before they affect daily operations.

04 BUSINESS ANALYTICS SUMMARY

4.1 Products Carried Across All or Most Stores

The following brands appear consistently across all or most of the 27 Burlington stores, indicating high consumer demand and strong OCS wholesale distribution. These form the baseline SKU set every store must carry competitively.

Brand / Product	Category	Burlington Ubiquity	Why It Sells
SHRED (various formats)	Flower / Pre-Roll	All 27 stores	Value positioning, bold branding, chain loyalty anchor. OCS volume leader.
Good Supply	Flower / Pre-Roll	All 27 stores	Accessible price point, reliable quality. Consistent OCS bestseller.
Redecan / Redees	Flower / Pre-Roll	All 27 stores	Ontario heritage brand. Value multipacks drive high unit velocity.
Spinach (Sourz, FEELZ)	Edibles / Vapes	All 27 stores	Flavour-forward, broad demographic appeal. Edibles staple.
Pure Sunfarms	Flower / Pre-Roll	24+ stores	Consistent quality and brand trust. Volume leader on OCS.ca.
Back Forty	Pre-Roll / Vapes	22+ stores	Value multipacks; rising vape brand with strong live resin entry pricing.
General Admission	Pre-Roll / Vapes	22+ stores	#1 Ontario pre-roll brand (May–Aug 2025, per OCS data).
WYLD	Edibles	18+ stores	Premium gummies with strong consumer repeat-purchase loyalty.
Simply Bare	Flower	16+ stores	Premium craft positioning. Enthusiast and wellness segment.

4.2 Sales Velocity & Sell-Through Potential by Category

Ontario's cannabis market generated \$2.1 billion in retail sales in 2024, with \$160M+ per month as of May 2025. Category velocity below is based on OCS published data and industry reporting:

Category	Ontario Share	Velocity	Strategic Insight
Pre-Rolls	~33%	VERY HIGH ↑	Surpassed flower in 2025. Infused formats are the fastest-growing sub-category in Canada.
Flower	~31%	HIGH →	Stable to slight decline. Craft and fresh-harvest differentiation growing among connoisseurs.
Vapes	~17%	HIGH ↑	Consistent growth. Live resin/rosin commands \$10–15 premium over distillate.

Category	Ontario Share	Velocity	Strategic Insight
Edibles	~9%	MEDIUM-HIGH ↑	100mg multipacks (introduced H2 2025) surging. Single 10mg units declining in relative share.
Concentrates	~6%	MEDIUM ↑	Rosin and live resin driving new interest. High-loyalty enthusiast segment.
Beverages	~2%	LOW-MEDIUM ↑	Fast-acting THC formats gaining. March 2026 THC+caffeine launch is a major catalyst.

4.3 Price Change Frequency

Price volatility in Burlington is primarily driven by four patterns:

- **Chain promotional cycles - Canna Cabana, Value Buds, and Hunny Pot rotate flash sales every 1–2 weeks. Flagship SKU prices can drop 15–25% for 3–7 day windows.**
- OCS wholesale pricing updates - OCS adjusts producer pricing periodically. A major pricing structure change in 2025 compressed margins on volume categories across all retailers.
- Clearance rotations - products approaching best-before dates see 20–40% mark-downs, most common in edibles and beverages.
- Seasonal events - 4/20, Cyber Monday, and holiday periods drive store-wide 10–20% promotions across all store types.

Key action: Daily HiBuddy price checks on the top 25 competitive SKUs vs. the 5 nearest competitors is the minimum monitoring cadence. The automated pipeline in Section 5 handles this at scale.

4.4 New & Upcoming OCS Products - Q1 2026 Launch Tracker

Product	OCS Launch	Strategic Relevance
THC + Caffeine Beverage (10mg THC / 30mg caffeine)	Mar 2026	Crossover appeal for active/wellness consumers. Novel format with broad demographic reach.
Afghan Double Black Hash-Infused Pre-Rolls	Mar 2026	Premium hash culture format. Strong demand signal from r/OntarioCannabis.
Pink Kush Live Resin Concentrate	Mar 2026	Indica favourite. Strong brand loyalty. High enthusiast demand.
THC:CBD:CBC Gummies 10mg/10mg/20mg	Mar 2026	Minor cannabinoid trend. CBC emerging as sought-after among educated consumers.
Transdermal THC:CBD Spray (1mg/spray)	Mar 2026	Pharmaceutical-grade wellness format. Differentiated from standard edibles.
Variable Temp USB-C Rechargeable Vape (THC+CBG+CBD)	Feb 2026	Multi-cannabinoid premium vape. Tech-forward appeal.
Simply Bare Live Rosin Softgels (10mg THC)	Feb 2026	Premium smokeless format. Appeals to consumers avoiding combustion.

Product	OCS Launch	Strategic Relevance
Clickstix Modular Snap-Together Vape System	Jan 2026	Major format innovation. Shareable and customizable flavour combinations.
Bubble Hash Live Rosin Pre-Roll with Glass Tip	Jan 2026	Solventless premium positioning. Strong quality signal for connoisseurs.
THC Fast-Act Tablets (10mg, breakable to 5mg)	Jan 2026	Precision dosing. Social consumption format for micro-dosers.

4.5 Consumer Sentiment - Reddit, Leafly & Review Analysis

Based on analysis of r/OntarioCannabis, r/canadiens, Google Reviews, Leafly, and Weedmaps for Burlington stores, the following themes emerge consistently:

- **Knowledgeable budtenders are the #1 reason cited in positive reviews for independent stores - the primary differentiator over chains.**
- Inventory gaps are the #1 trigger for negative reviews. Consumers who find a product out of stock rarely return.
- Price-shopping is intense. Consumers openly share price comparisons on Reddit and use HiBuddy before leaving home.
- Infused pre-rolls generate the highest excitement on Reddit - hash hole, diamond-infused, and rosin formats discussed weekly.
- 100mg edible multipacks have strong positive sentiment since their H2 2025 introduction - consumers see major value over single 10mg units.
- Independent stores (PEPE, BUDSSMOKE, Hidden Leaf) are praised for curating unique SKUs not available at chains - a positioning advantage MontKailash shares.
- Oakville proximity is a recurring theme - Burlington stores are the only licensed option for Oakville residents, driving meaningful cross-boundary traffic.

05 DATA ENGINEERING PIPELINE ARCHITECTURE

This section defines the end-to-end automated data pipeline that replaces manual, ad-hoc research with a fully automated, observable, and scalable system delivering daily competitive intelligence to MontKailash Cannabis. The pipeline follows a modern ELT (Extract, Load, Transform) pattern.

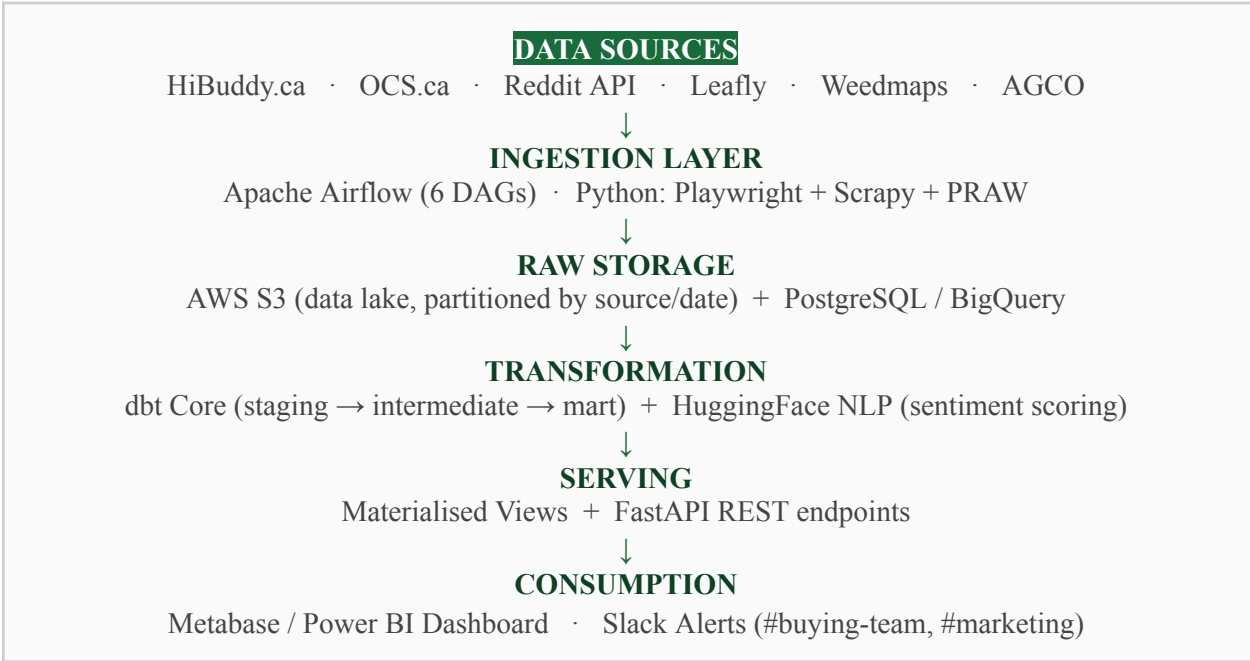
5.1 Architecture Overview

The pipeline has six logical layers, each independently scalable and observable:

Layer	Primary Tools	Output
Layer 0 - Data Sources	HiBuddy.ca, OCS.ca, Reddit API, Leafly, Weedmaps, AGCO	Raw HTML / JSON / API responses
Layer 1 - Ingestion	Apache Airflow + Python (Playwright, Scrapy, PRAW)	Validated JSON → S3 raw bucket
Layer 2 - Raw Storage	AWS S3 (data lake) + PostgreSQL / BigQuery	Partitioned parquet files; raw SQL tables
Layer 3 - Transformation	dbt Core + spaCy / HuggingFace (NLP)	Mart tables: price_delta, availability, sentiment
Layer 4 - Serving	Materialised views + FastAPI REST API	BI-ready tables; REST endpoints
Layer 5 - Consumption	Metabase / Power BI dashboard + Slack alerts	Daily KPI dashboards; automated price-drop notifications

5.2 Pipeline Diagram

The diagram below illustrates the end-to-end data flow from source collection through to business consumption:



5.3 Ingestion - Airflow DAG Schedule

Each data source runs as a separate Airflow DAG, scheduled independently to match source update frequency. DAG-level SLA monitoring triggers engineering alerts on any missed run.

DAG Name	Schedule	Description
dag_hibuddy_menus	Daily 06:00 ET	Scrapes all 27+ Burlington store menus and prices from HiBuddy.ca (Playwright). Full product snapshot stored to S3.
dag_ocs_products	Daily 07:00 ET	Polls OCS.ca new-products page and OCS Wholesale feed. Flags new SKUs in categories MontKailash carries.
dag_reddit_sentiment	Daily 08:00 ET	Pulls last 24h posts from r/OntarioCannabis and r/canadiens via PRAW API. Keyword-filtered for Burlington, store names, and brands.
dag_leafly_weedmaps	Every 48h	Scrapes Burlington store ratings, review counts, and deal listings from Leafly and Weedmaps.
dag_store_details	Weekly (Sunday)	Crawls individual store websites for hours, phone, address changes. Feeds dim_stores SCD Type 2 updates.
dag_price_alerts	Post dag_hibuddy	Compares today vs. yesterday prices. Fires Slack alert to #buying-team on any SKU drop exceeding 10%.

All scrapers use retry logic (tenacity), rotating proxy pools to avoid rate limits, and randomised user-agent headers. HiBuddy.ca requires Playwright (headless Chromium) because menus are JavaScript-rendered - static HTTP scraping will not capture dynamic product data.

5.4 Storage - Database Schema

Raw data lands in AWS S3 first - partitioned by source and date. This is the system of record; if a downstream bug corrupts transformed data, raw files are always available for full reprocessing (24-month retention). Structured data is then loaded into PostgreSQL or BigQuery.

Table	Grain	Key Columns
dim_stores	One row per store	store_id, name, address, phone, type, latitude, longitude, agco_licence, valid_from (SCD Type 2)
dim_products	One row per SKU	product_id, name, brand, category, ocs_id, the_pct, cbd_pct, weight_g
fct_prices	Store × SKU × date	store_id, product_id, scraped_date, regular_price, sale_price, in_stock
fct_price_history	Append-only (immutable)	All columns from fct_prices - never updated, enabling point-in-time price queries
fct_sentiment	Reddit post level	post_id, subreddit, brands_mentioned[], sentiment_score, confidence, created_utc

Table	Grain	Key Columns
fct_ocs_launches	OCS product × launch date	product_id, launch_date, name, brand, category, ocs_price, url

fct_price_history is append-only and immutable - enabling accurate point-in-time price queries and historical trend analysis. dim_stores uses SCD Type 2 so that address or hours changes are preserved as history rather than overwritten.

5.5 Transformation - dbt Model Layer

dbt Core manages all SQL transformations across three layers (staging → intermediate → mart), all version-controlled in Git with automated tests run on every code change via GitHub Actions CI/CD.

dbt Model	Layer	Purpose
stg_hibuddy_products	Staging	Normalises raw HiBuddy JSON into clean product records
stg_ocs_products	Staging	Normalises OCS product feed with launch date and price
stg_reddit_posts	Staging	Cleans Reddit posts; extracts keyword matches
int_price_with_history	Intermediate	Joins current price with prior-day snapshot to compute delta
int_sku_coverage	Intermediate	SKU availability matrix: which stores carry each product
mart_price_delta	Mart	Daily % price change per SKU per store. Flags drops >10% for alert trigger.
mart_competitor_gaps	Mart	Products at 3+ competitors that MontKailash doesn't carry - ranked by ubiquity. Direct daily buying signal.
mart_product_velocity	Mart	Proxy sell-through metrics: availability gaps, restock frequency, cross-store demand
mart_ocs_new_arrivals	Mart	New OCS products <30 days old, filtered to MontKailash's active categories
mart_sentiment_trends	Mart	Brand-level sentiment over time: positive/negative/neutral ratio + weekly mention counts

mart_competitor_gaps is the highest-value buying signal: it identifies every product stocked at 3+ competitor stores that MontKailash does not carry, ranked by competitor ubiquity - producing a daily automated recommended buy list for the purchasing team.

5.6 Consumption - Dashboard & Alerts

Business outputs are delivered via Metabase (recommended over Power BI for cost and ease of use by non-technical staff). Six dashboard pages cover the full operational picture:

- Market Overview - store count by type, category market share, MontKailash price position vs. market average

- Daily Price Tracker - price delta table, SKUs on sale today, competitor flash deal calendar
- Competitor Gaps - products at 5+ competitors that MontKailash doesn't carry; automated recommended buy list
- OCS New Arrivals - all new OCS products in the last 30 days, category breakdown, order status tracker
- Product Velocity - category velocity bars, top 20 SKUs by estimated sell-through, out-of-stock frequency
- Sentiment Monitor - brand sentiment scores, Reddit post volume, trending keywords and discussions

Slack alerts are triggered by the dag_price_alerts DAG and fire to the appropriate team channel when:

- A competitor drops any SKU MontKailash carries by more than 10% vs. the previous day - fires to #buying-team immediately after the morning scrape.
- A new OCS product appears in a category MontKailash actively stocks - fires daily at 07:30 ET.
- A product is newly stocked at 3+ competitors that MontKailash does not carry - fires to #buying-team at 09:00 ET.
- A brand or product mention volume spikes more than 2 standard deviations above the 14-day average - fires to #marketing.

5.7 Technology Stack & Rationale

The following tools were selected for reliability, low operational overhead, and suitability for a small team:

Orchestration: Apache Airflow (AWS MWAA) - industry-standard scheduling; managed version eliminates infra ops overhead

Web Scraping: Playwright (JS-rendered pages: HiBuddy, Leafly) + Scrapy (static HTML: OCS) - both support proxy rotation and retry

Reddit API: PRAW (Python Reddit API Wrapper) - official client, rate-limit aware, handles auth and pagination automatically

Raw Storage: AWS S3 - cheap, durable, infinitely scalable object storage; partitioned by source and date

Data Warehouse: PostgreSQL (RDS) for smaller scale / lower cost; BigQuery if analytics volume grows - both integrate natively with dbt

Transformation: dbt Core - SQL-first, version-controlled, testable; free open-source; CI/CD via GitHub Actions

NLP / Sentiment: HuggingFace transformers + spaCy - pre-trained sentiment models fine-tuned on cannabis review corpus; spaCy for entity extraction

Serving API: FastAPI + Uvicorn - lightweight async Python API; auto-generates OpenAPI docs; enables future mobile/POS integration

BI Dashboard: Metabase (self-hosted) - open-source; non-technical users query without SQL; daily-refresh for exec and store management

Alerting: Slack Webhooks - zero-friction push notifications to existing team channels; no new tooling for end users

Version Control: GitHub + GitHub Actions - all pipeline code, dbt models, and DAG definitions in Git; dbt tests run on every PR

Containerisation: Docker + Docker Compose - reproducible local dev environment; deploy to EC2 or ECS in production with no drift

Secrets: AWS Secrets Manager - API keys, DB credentials, Slack webhook URLs never stored in code

5.8 Infrastructure Cost Estimate

Service	Tier	Monthly Cost (USD)
AWS MWAA (Airflow)	Small environment	~\$75–\$120 / month
AWS RDS PostgreSQL	db.t3.small	~\$25–\$40 / month
AWS S3 raw data lake	~50 GB/month	~\$2–\$5 / month
EC2 (scrapers + FastAPI)	t3.small	~\$15–\$20 / month
Metabase (self-hosted on EC2)	t3.micro	~\$10 / month
GitHub, dbt Core, Slack	Free tiers	\$0
TOTAL	—	~\$130–\$200 USD / month

Costs assume small-scale deployment for the Burlington market (27 stores, daily refresh). A fully managed Airflow alternative (Astronomer Cloud) runs ~\$200–300/month but eliminates all infrastructure management - suitable if team capacity is limited.

5.9 Data Quality & Observability

All dbt models include automated tests: not_null constraints, referential integrity checks, accepted value ranges for price fields, and row-count minimums per daily scrape run.

Additional monitoring: if fct_prices is not updated by 10:00 ET, an alert fires to the engineering Slack channel. Each daily scrape is expected to return data for at least 80% of the 27-store census - a lower count signals a site change or blocking event before it affects the buying team.

5.10 Implementation Roadmap

Phase	Timeline	Deliverables
Phase 1	Weeks 1–2	HiBuddy Playwright scraper + S3 raw lake + PostgreSQL schema + Airflow DAG skeleton
Phase 2	Weeks 3–4	OCS scraper + Reddit PRAW collector + staging dbt models + automated data quality tests
Phase 3	Weeks 5–6	Mart dbt models (price_delta, competitor_gaps, velocity) + Metabase dashboard v1
Phase 4	Weeks 7–8	NLP sentiment pipeline + Slack alert system + FastAPI REST endpoints
Phase 5	Weeks 9–10	Full observability (CloudWatch, dbt source freshness) + documentation + team handover

06 STRATEGIC RECOMMENDATIONS

6.1 Product Strategy - Win on Curation

- **Stock the March 2026 OCS drops immediately, especially the THC+caffeine beverage and hash-infused pre-rolls. Being first-to-shelf on trending new products generates foot traffic and social media commentary.**
- Build a 'MontKailash Picks' shelf - a curated 15–20 SKU featured section updated monthly, highlighting new OCS launches and craft producers.
- Prioritize infused pre-rolls (General Admission, BOXHOT, Jeeter, diamond and hash variants) - the fastest-growing format in Canada. Maintain deep stock depth.
- Stock 100mg edible multipacks from multiple brands - consumer demand surged in H2 2025; this is now a core SKU tier, not optional.
- Maintain craft flower: Simply Bare, Ghost Drops, Qwest, artisan brands. This is the clearest differentiator from chains and builds high-loyalty repeat customers.
- Monitor minor cannabinoid products (CBG, CBN, CBC formulations) - early adoption signals credibility to the most educated and high-spending consumers.

6.2 Pricing Strategy - Smart Parity, Not a Race to the Bottom

- **Monitor HiBuddy daily for the top 25 competitive SKUs across the 5 nearest Canna Cabana, Hunny Pot, and One Plant locations. Match pricing within \$1–2 on high-velocity items.**
- Differentiate on value-adds over pure price: loyalty points, a free pre-roll with \$80+ purchase, senior discounts, and first-visit discounts rather than blanket markdowns.
- Hold premium price points on craft and specialty products - consumers buying Ghost Drops or Simply Bare are not chain shoppers.
- Implement a weekly 'Staff Pick' rotating deal (1–2 products, 15–20% off for 7 days) to drive repeat traffic without undermining overall margin structure.

6.3 Operations - Inventory & Digital Presence

- **Maintain a minimum 150+ active SKUs at all times. Inventory gaps are the #1 negative review trigger and permanently lost customers.**
- Use the OCS Wholesale dashboard to pre-order new launches 2–3 weeks in advance - authorized retailers see upcoming products before public announcement.
- Keep the HiBuddy menu updated daily with accurate pricing and in-stock status. HiBuddy is the primary discovery channel for price-conscious consumers.
- Respond to all Google Reviews within 48 hours - active review management is consistently cited in positive store assessments across Burlington.

6.4 Brand Positioning - Independent Advantage

- **Lean into the 'independent, locally owned' identity. Reddit and review data consistently shows Burlington consumers choose independents for service and curation - MontKailash's structural advantage over Canna Cabana and Hunny Pot.**
- Engage the Oakville spillover market actively - zero stores in Oakville creates sustained cross-boundary demand. Google Ads geotargeting Oakville postal codes is a low-cost, high-return channel.

- Train staff intensively and visibly on terpene profiles, cannabinoid ratios, and new product education. Staff knowledge is the #1 cited reason consumers choose an independent over a chain.

6.5 Ongoing Competitive Monitoring Schedule

The data pipeline in Section 5 automates the daily and weekly tasks below. Manual oversight responsibilities:

Activity	Source	Frequency	Owner
Price check - top 25 SKUs vs. 5 nearest competitors	HiBuddy.ca	Daily	Operations Manager
New OCS product review and pre-order planning	OCS Wholesale	Monthly (1st)	Buyer / Owner
Consumer review monitoring	Google, Leafly, Weedmaps	Weekly	Store Manager
Reddit sentiment scan	r/OntarioCannabis, r/canadians	Weekly	Marketing
AGCO new store applications - Burlington area	AGCO iAGCO portal	Monthly	Owner

07 ASSUMPTIONS & DATA LIMITATIONS

7.1 Stated Assumptions

- **Store count: The 27-store Burlington total is from HiBuddy.ca, cross-referenced with AGCO, Weedmaps, and Leafly as of March 2026. A small number may have opened post-collection.**
- Hours of operation: Sourced from store websites, Google Business Profiles, and HiBuddy.ca. Standard hours only - holiday and temporary adjustments may differ. AGCO mandates maximum 9 AM–11 PM.
- Pricing: Reflects market-representative data as of March 2026. OCS is the sole legal wholesaler; all retailers pay the same wholesale price. Retail variation reflects margin decisions and promotional strategy.
- Owner details: AGCO does not bulk-disclose full operator details. Individual CRSA documents require a formal iAGCO records request. Publicly confirmed names are noted where available.
- 35 km radius count: The ~60–70 estimate is directional. An exact count requires a formal AGCO radius-based authorization search.
- Sentiment analysis: Reddit and review sentiment is directional, not statistically definitive. Volume of Burlington-specific posts is moderate; insights are indicative.

7.2 Recommended Ongoing Data Infrastructure

- Subscribe to HiBuddy for Business for automated price and menu tracking - eliminates manual daily monitoring.
- Integrate OCS Wholesale dashboard for forward-looking product planning with 30-day advance notice on upcoming drops.
- Set up Google Alerts for AGCO new store authorization announcements in Burlington and Halton Region.
- Consider BudSense or a comparable POS analytics platform for Ontario retail benchmarking and real sell-through data.
- The data engineering pipeline in Section 5 automates all of the above at scale with daily refresh, full price history, and proactive Slack alerts.

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*Sources: AGCO · OCS.ca · HiBuddy.ca · Weedmaps · Leafly · Cannasearch.ca · ADCANN · StratCann ·
Reddit · Apache Airflow · dbt Core · AWS · HuggingFace*